

NASA Says Spacecraft Made Closest-Ever Pass to Sun

NASA says a spacecraft that has been examining the sun's outermost atmosphere recently got closer to our Earth's star than any past explorer has.

The American space agency announced its Parker Solar Probe passed within 6.1 million kilometers of the sun's outer corona on December 24. Mission leaders established communication with the spacecraft after the operation and said the orbiter was safe.

NASA launched the Parker Solar Probe in August 2018. Its mission is to study the sun's powerful corona and solar wind. Solar wind is created by a continuous flow of charged particles, called plasma, into space from the corona.

In 2021, NASA announced the orbiter had passed into the solar atmosphere for the first time. Researchers said at the time, the spacecraft had successfully "touched" the sun. Parker was believed to be about 13 million kilometers from the center of the sun when it crossed over into the sun's outer atmosphere.

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In a statement, NASA's Associate Administrator Nicky Fox praised the spacecraft's latest pass-by of the sun. "Flying this close to the sun is a historic moment in humanity's first mission to a star," she said. Fox leads the agency's Science Mission Directorate at NASA Headquarters in Washington, D.C.

The Parker Solar Probe is believed to have traveled at speeds up to 692,000 kilometers per hour during the mission. NASA said this was "faster than any human-made object has ever moved."

Fox said she hopes data collected by the spacecraft can help scientists better understand how the sun behaves and influences other elements in our solar system and beyond. "By studying the sun up-close, we can better understand its impacts throughout our solar system, including on the technology we use daily on Earth and in space."

Fox added that such studies can provide valuable information "about the workings of stars across the universe to aid in our search for habitable worlds beyond our home planet."

This artist's impression depicts the Parker Solar Probe on a flight around the Sun. (Image Credit: NASA/Johns Hopkins APL/Steve Gribben)

NASA's statement noted the agency's earlier preparation and operations were all leading up to the latest mission. To help it get into the right position, the Parker Solar Probe received

assistance from several flybys of the planet Venus. The last flyby happened on November 6, 2024.

These gravitational assists had put the spacecraft “an ideal distance” from the sun every three months, NASA explained. This distance was “close enough to study our sun’s mysterious processes,” but not “to become overwhelmed by the sun’s heat and damaging radiation.”

Developers of the spacecraft say it was built with a shield made with carbon material to protect it from the extreme heat released by the sun’s corona. The shield was designed to survive temperatures reaching 1,400 degrees Celsius. NASA says the data-gathering instruments are placed behind the shield.

Nour Rawafi is the project scientist for Parker Solar Probe at NASA’s Johns Hopkins Applied Physics Laboratory in Maryland. His team designed, built, and operates the spacecraft. He said Parker had performed as planned although it operates in “one of the most extreme environments in space.”

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Rawafi added that the mission had brought “a new golden era of space exploration, bringing us closer than ever to unlocking the sun’s deepest and most enduring mysteries.”

Dr. Joseph Westlake is director of NASA’s Heliophysics Division, the part of the agency that studies the sun. Westlake noted that his team is preparing for more flybys in the future and hopes to capture more valuable data. Already, flybys of Venus have gathered information about the natural radio waves coming from Venus.

In 2023, scientists announced they had learned new information about solar wind from data collected by Parker. Data showed that solar wind can reach speeds up to 1.6 million kilometers per hour. Solar wind forms a large magnetic barrier known as the heliosphere. It protects Earth and other planets from the continuous flow of high-energy particles from the sun’s corona.

I’m Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, Agence France-Presse and NASA.

Quiz - NASA Says Spacecraft Made Closest-Ever Pass to Sun

Vocabulary

corona - n. ชั้นบรรยากาศภายนอกของดาวฤกษ์ เช่น ดวงอาทิตย์

impact - n. ผลกระทบที่บุคคล เหตุการณ์ หรือสถานการณ์มีต่อบางสิ่งบางอย่าง

habitable - adj. สามารถรองรับสิ่งมีชีวิตได้

flyby - n. คำที่หน่วยงานอวกาศใช้เพื่ออธิบายเมื่อยานอวกาศใช้แรงโน้มถ่วงของดาวเคราะห์เพื่อเพิ่มความเร็วหรือเปลี่ยนทิศทาง

overwhelm - v. ทำให้รู้สึกท่วมท้น หรือได้รับผลกระทบมากเกินไป

shield - n. วัตถุขนาดใหญ่และแบน ใช้สำหรับป้องกันอันตรายหรือไม่พึงประสงค์

golden era - n. ยุคทอง ช่วงเวลาที่บางสิ่งบางอย่างหรือใครบางคนประสบความสำเร็จเป็นพิเศษ ดีเยี่ยม หรือพิเศษ

endure - v. อดทนผ่านช่วงเวลาหรือสถานการณ์ที่ยากลำบาก

heliophysics - n. เกี่ยวกับพื้นที่ในอวกาศที่ได้รับอิทธิพลจากดวงอาทิตย์หรือลมสุริยะ

delve - v. เจาะลึก ตรวจสอบสิ่งใดสิ่งหนึ่งโดยละเอียดเพื่อค้นหาข้อมูล

Reading Comprehension Quiz

1. What significant achievement of a NASA spacecraft is highlighted in the article's opening?

It discovered a new star closer to Earth.

It successfully launched a new type of satellite.

It got closer to the sun than any previous explorer.

It completed a full orbit around the sun for the first time.

2. When did the Parker Solar Probe make its most recent close pass to the sun's outer corona, and at what distance?

November 6, 2024, at 13 million kilometers.

August 2018, at 6.1 million kilometers.

December 24, at 6.1 million kilometers.

In 2021, at 13 million kilometers.

3. What is the primary mission of the Parker Solar Probe, which was launched in August 2018?

To photograph distant galaxies.

To study the sun's powerful corona and solar wind.

To search for new planets in our solar system.

To test new propulsion technologies for spacecraft.

4. According to the article, how is solar wind created?

By the intense gravitational pull of the sun's core.

From a continuous flow of charged particles (plasma) from the corona.

Through massive solar flares and eruptions.

By the interaction of sunlight with planetary atmospheres.

5. In what year did the Parker Solar Probe first 'touch' the sun's atmosphere, and approximately how far was it from the sun's center at that time?

August 2018, about 6.1 million kilometers.

December 24, about 13 million kilometers.

2021, about 13 million kilometers.

November 6, 2024, about 6.1 million kilometers.

6. Who is Nicky Fox, and what was her description of the spacecraft's latest pass-by of the sun?

She is the mission leader, calling it a routine operation.

She is an astronaut, calling it a dangerous maneuver.

She is NASA's Associate Administrator, calling it a historic moment.

She is a lead scientist, calling it a minor technical achievement.

7. During its mission, what speed record did the Parker Solar Probe reportedly break?

The fastest speed for an orbiting satellite around Earth.

The fastest speed ever achieved by a human-made object.

The fastest speed for a probe entering a star's atmosphere.

The fastest speed for data transmission from deep space.

8. According to Nicky Fox, how can studying the sun up-close specifically benefit us on Earth?

It helps us predict terrestrial weather patterns more accurately.

It provides new sources of renewable energy.

It helps us better understand its impacts on technology we use daily.

It reveals the composition of the sun's core directly.

9. Beyond understanding our solar system, what other potential benefit does Nicky Fox mention that these studies could provide?

Improved methods for interstellar travel.

Valuable information about the workings of stars to aid in the search for habitable worlds.

New theories on the origin of the universe.

A deeper understanding of black holes and distant galaxies.

10. How did the Parker Solar Probe achieve the correct position for its mission, and when was its last assisting maneuver?

By using powerful onboard thrusters in 2023.

Through several flybys of Mars, the last one in early 2024.

With assistance from several flybys of Venus, the last one on November 6, 2024.

By launching directly into its final orbit without external assistance.

11. What was the main purpose of the gravitational assists (flybys) from Venus?

To gather data specifically about Venus's atmosphere.

To put the spacecraft at an ideal distance from the sun every three months.

To test the spacecraft's durability against intense radiation.

To refuel the spacecraft for extended operations.

12. What material is the Parker Solar Probe's shield made of, and what temperature is it designed to survive?

Titanium, up to 1,000 degrees Celsius.

Carbon, up to 1,400 degrees Celsius.

Ceramic, up to 1,200 degrees Celsius.

Steel alloy, up to 900 degrees Celsius.

13. Who is Nour Rawafi, and what was his overall assessment of the Parker Solar Probe's performance?

He is the lead engineer, stating the probe exceeded expectations despite minor issues.

He is the project scientist, stating the probe performed as planned despite extreme conditions.

He is the mission director, stating the probe experienced unexpected malfunctions.

He is a data analyst, stating the data collected was incomplete.

14. According to Nour Rawafi, what broader impact has the Parker Solar Probe mission brought to space exploration?

It marked the end of an era of close-sun studies.

It brought a new golden era, unlocking the sun's deepest mysteries.

It primarily focused on developing new materials for spacecraft.

It confirmed existing theories without significant new discoveries.

15. What is the function of the heliosphere, which is formed by solar wind?

It absorbs harmful ultraviolet radiation from the sun.

It generates the sun's magnetic field.

It protects Earth and other planets from high-energy particles from the sun's corona.

It regulates the sun's temperature and energy output.